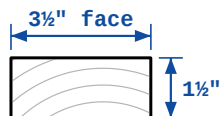


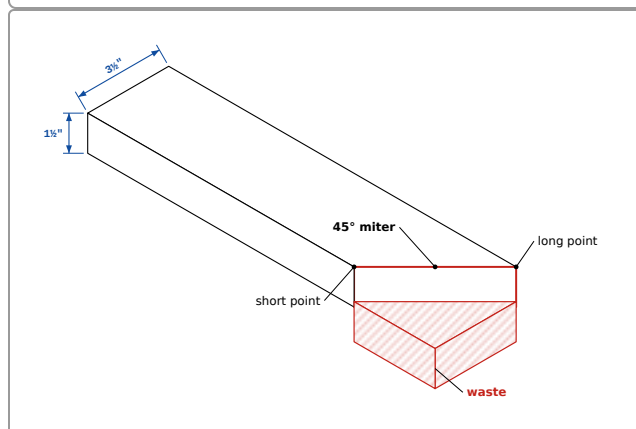
THE ANGLED-CUT HANDBOOK

How to work out the angle, the length, and the saw setting for any sloped or mitered cut, then mark it, cut it, and check it. Plain language first, then the formula, then real numbers.

 Board
  Waste (cut away)
  Cut line
  Dimension
  Angle



Running example: a 2×4. It is sold as a “2 by 4” but actually measures $1\frac{1}{2}'' \times 3\frac{1}{2}''$. The wide side is the **face** ($3\frac{1}{2}''$), the narrow side is the **edge** ($1\frac{1}{2}''$). We use those real numbers all the way through.



A 2×4 with a 45° miter. The hatched wedge is the waste the saw removes.

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THE VOCABULARY OF ANGLED CUTS

Crosscut · rip · miter · bevel · compound · long point · short point · kerf

1.1 Crosscut vs rip cut

Wood has **grain**: long fibres that run the length of the board, like a bundle of straws. Every cut is named by how it crosses those fibres.

- A **crosscut** goes across the grain. It makes a board *shorter*. Every angled cut in this guide is a kind of crosscut.
- A **rip cut** goes along the grain. It makes a board *narrower*.

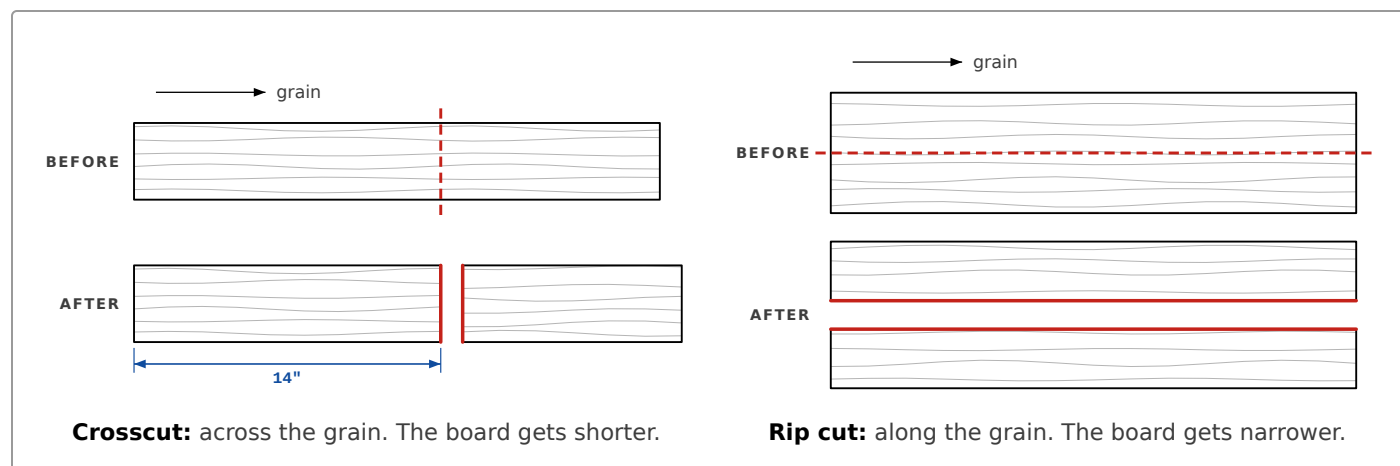


FIG 1.1 Top views, before and after. The grain lines show which way the fibres run.

COMMON MISTAKE

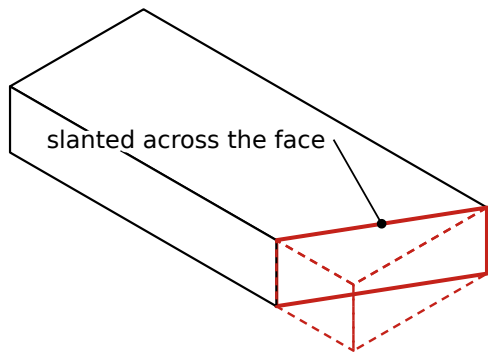
Using a rip blade for crosscuts (or the reverse). Rip blades have few, large teeth and tear the fibres when crosscutting, which leaves a ragged miter. For clean angled cuts, use a crosscut or combination blade with more teeth.

1.2 Miter, bevel, and compound

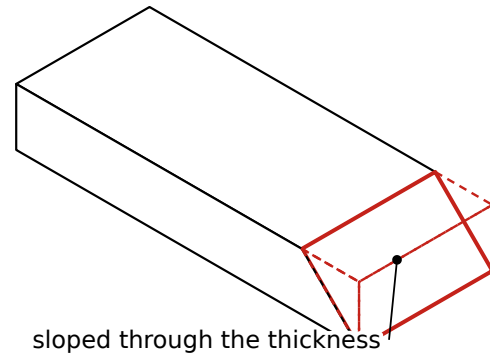
A plain crosscut is square: straight across, at 90° to the edges. An angled crosscut leans in one of two directions, or both.

- A **miter** is angled across the *face*. Looking down at the board, the end is slanted. Picture frames use miters.
- A **bevel** is angled through the *thickness*. Looking down, the end looks square, but it slopes when you look at the edge.
- A **compound** cut is both at once: slanted across the face *and* sloped through the thickness. Crown molding and splayed boxes need these.

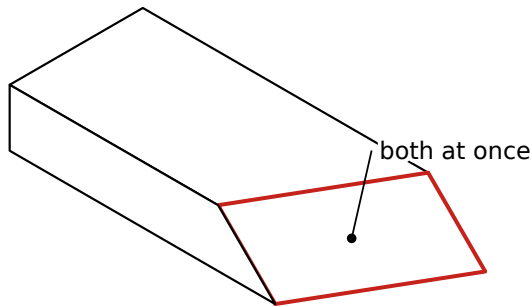
On a miter saw, the *miter* setting swings the saw left and right; the *bevel* setting tilts the blade over.



Miter · 30° across the face

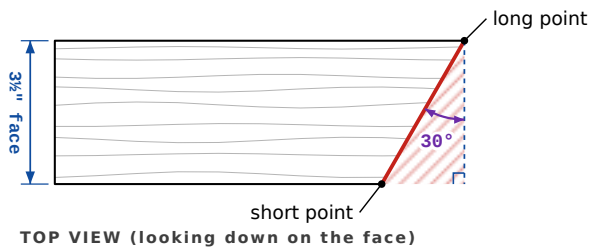


Bevel · 45° through the thickness

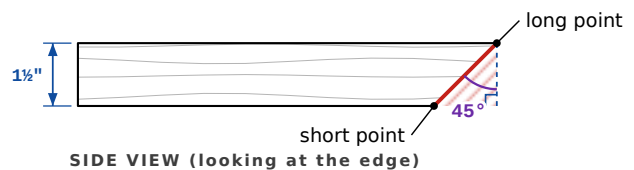


Compound · 30° miter + 45° bevel

FIG 1.2 The same 2×4 end cut three ways (isometric view). The red-outlined face is the fresh cut; the dashed outline is the wedge of waste that was removed.



Miter: the angle shows on the face.



Bevel: the angle shows on the edge.

FIG 1.3 Where each angle shows up. A miter is visible from above; a bevel is visible from the side. The angle is measured from a square line (dashed blue), the line a plain 90° cut would follow.

COMMON MISTAKE

Setting the bevel when you meant the miter. Both scales read 45° for a picture-frame corner, so the numbers look the same. Ask: “Should the slant show when I look down at the face (miter), or when I look at the edge (bevel)?”

1.3 Long point and short point

An angled cut makes one edge of the board longer than the other. The tip of the longer edge is the **long point** (the sharp corner). The tip of the shorter edge is the **short point** (the blunt corner, also called the heel).

Carpenters measure mitered parts “long point to long point” because the sharp tip is easy to hook a tape on and easy to see. On a frame, the long points sit on the outside edge and the short points on the inside edge.

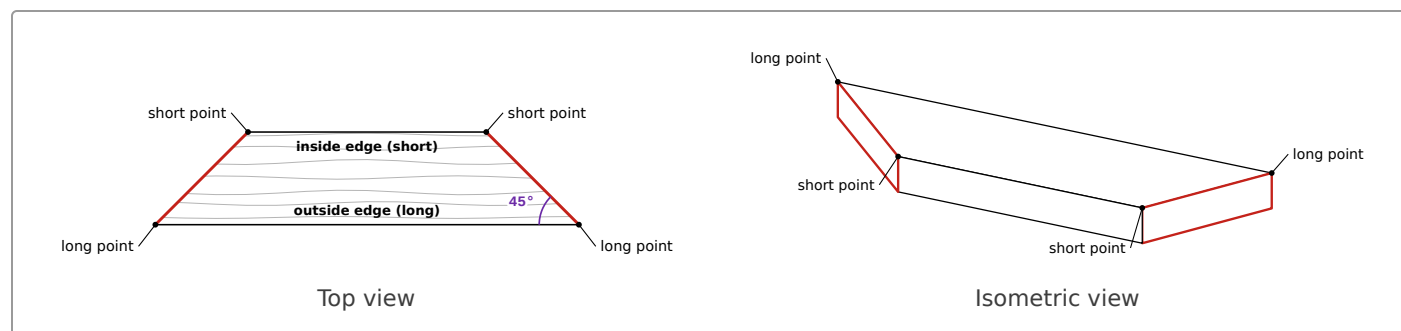


FIG 1.4 One side of a picture frame, mitered 45° at both ends. Top view on the left, isometric on the right.

COMMON MISTAKE

Writing “cut to 20 inches” on a cut list without saying which edge. A 2×4 mitered at 45° is 7” shorter on one edge than the other, so “20 inches” can mean two very different parts. Always write *LP* (long point) or *SP* (short point) next to the length.

1.4 Kerf: the wood the blade eats

A saw blade has thickness. The slot it leaves behind is the **kerf**, usually about $\frac{1}{8}$ ” for a full-size circular or miter saw blade (thin-kerf blades are around $\frac{3}{32}$ ”). That wood turns to sawdust, so it is gone from both pieces.

The rule that follows: **put the kerf on the waste side of your line.** The part you keep is called the **keeper**; the other part is the **waste** (or offcut).

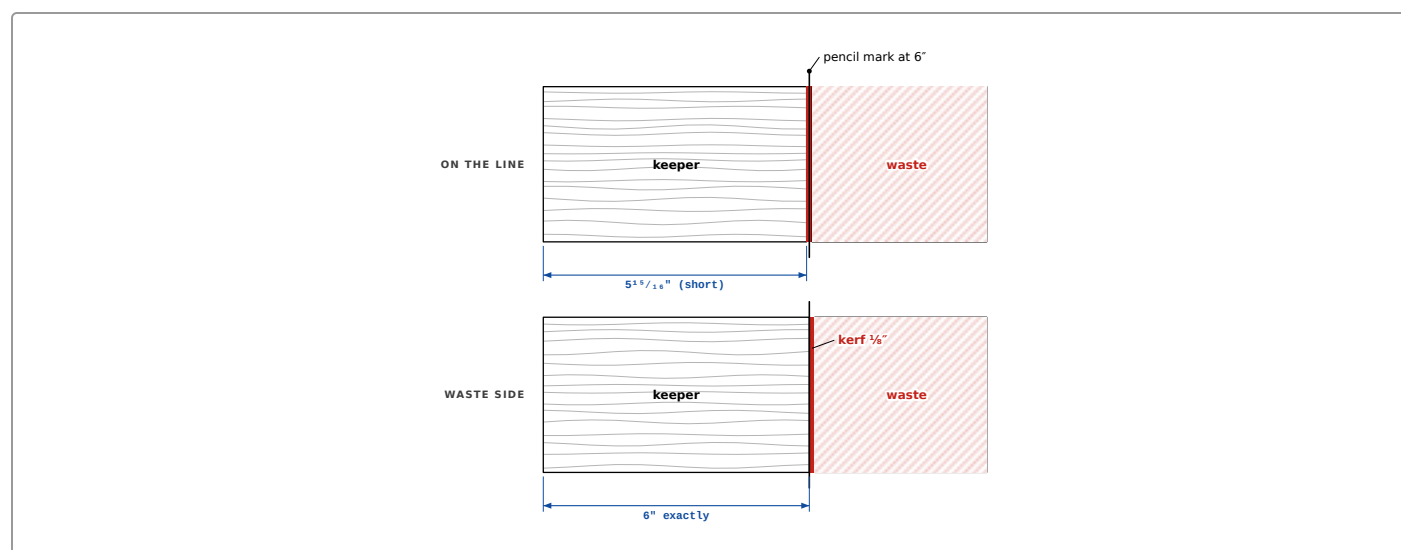


FIG 1.5 Drawn to scale with a $\frac{1}{8}$ ” kerf. Cutting down the middle of the pencil line shortens the keeper by half a kerf.

FORMULA

material used = $\Sigma(\text{piece lengths}) + (\text{number of cuts}) \times \text{kerf}$

Σ means “add them all up”.

WORKED EXAMPLE

You want four 18" pieces from an 8-foot (96") 2×4, cut with a $\frac{1}{8}$ " kerf.

1. Pieces: $4 \times 18" = 72"$

2. Cuts: 4 (the last one frees the fourth piece from the leftover): $4 \times \frac{1}{8}" = \frac{1}{2}"$

3. Total used: $72\frac{1}{2}"$

You use $72\frac{1}{2}"$ and have $23\frac{1}{2}"$ left over. Kerf only matters here when the board is tight: five 19" pieces ($95"$ plus five $\frac{1}{8}"$ kerfs = $95\frac{5}{8}"$) would *not* fit in a 96" board with room to square the end.

COMMON MISTAKE

Marking all the pieces along the board at 18", 36", 54", 72" and then cutting. Each cut eats into the next piece, so pieces 2, 3 and 4 come out short. Measure and mark one piece at a time from the freshly cut end.

GETTING THE RIGHT ANGLE FOR THE JOINT

The half-angle rule · polygon frames · out-of-square corners · braces and T-joints · saw scale vs geometry

2.1 The core rule: split the corner in half

When two boards meet at a corner and both are cut, the joint line runs exactly through the middle of the corner. The fancy word is **bisector**: the line that splits an angle into two equal halves. Because the joint is shared, each board takes half of the corner.

Intuition: fold a sheet of paper so one edge lies on the other. The crease is the bisector, and it is where the two boards touch.

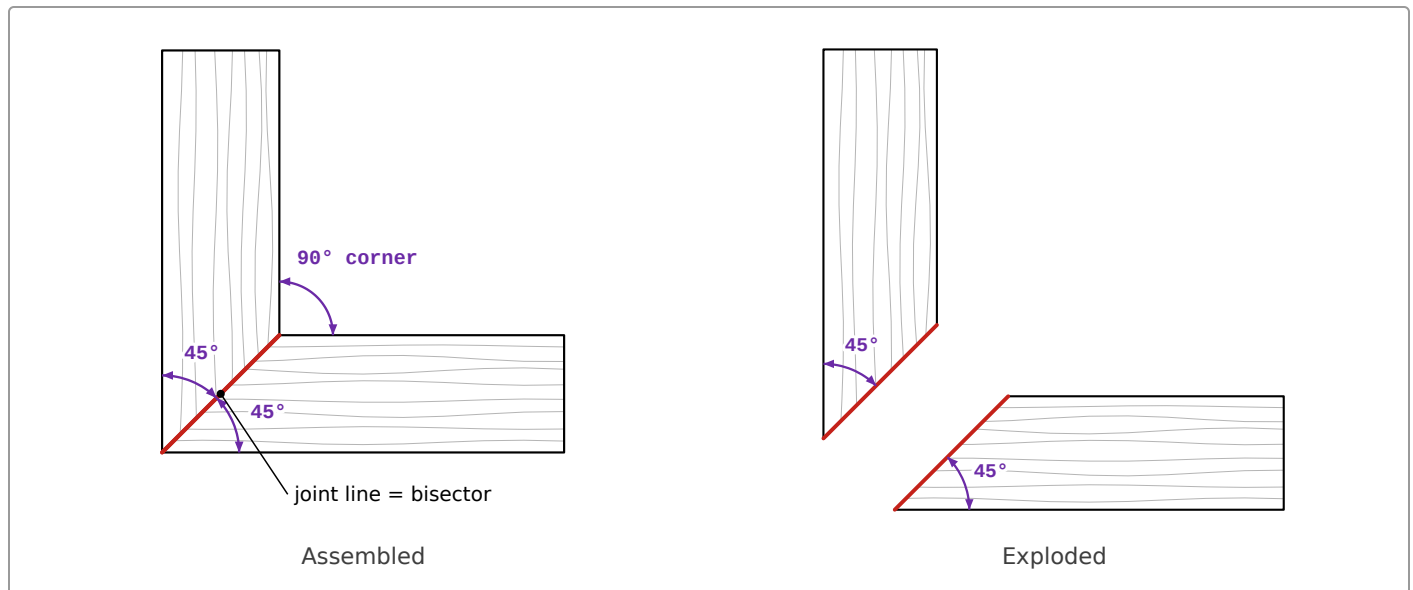


FIG 2.1 A 90° corner. Assembled (left), each piece carries a 45° angle between its cut and its edge. Pulled apart (right), you can see the two halves.

FORMULA

angle between cut and edge = $C \div 2$

saw setting = $90^\circ - C \div 2$

C is the corner angle: the angle between the two boards, measured on the side where they touch. The second line converts to a miter-saw reading, explained in 2.5.

WORKED EXAMPLE

A square picture frame has four 90° corners.

1. Angle between cut and edge: $90^\circ \div 2 = 45^\circ$
2. Saw setting: $90^\circ - 45^\circ = 45^\circ$

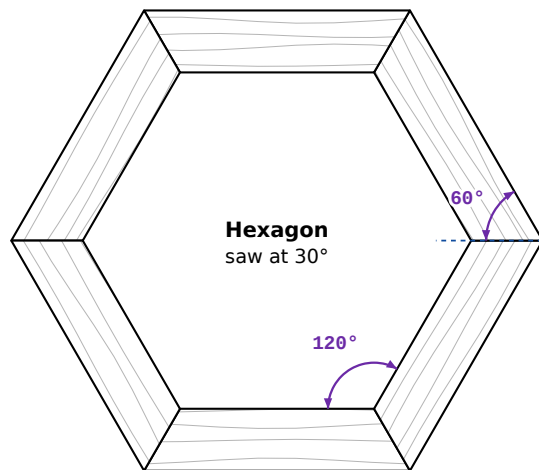
Every end gets a 45° miter. At 90° the two numbers happen to agree, which is why the saw's different scale goes unnoticed until you try another corner.

COMMON MISTAKE

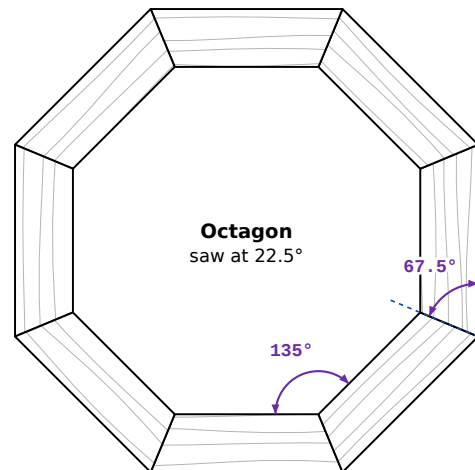
Cutting one piece at the full corner angle and leaving the other square. That only works for a *butt joint* where one board lands on the face of another (see 2.4). For a corner where both ends show, split the angle.

2.2 Polygon frames

A regular polygon (a shape with n equal sides) has n equal corners. Walking around its outline you turn a full 360° in total, so each corner turns $360^\circ \div n$. Each corner is shared by two cuts, so each cut takes half of that turn: $180^\circ \div n$. That is the saw setting directly.



$n = 6 \cdot$ corner $120^\circ \cdot$ saw 30°



$n = 8 \cdot$ corner $135^\circ \cdot$ saw 22.5°

FIG 2.2 A hexagon and an octagon frame made from 2x4s laid flat. Alternate pieces are shaded so you can see the joints. Purple arcs show the corner angle and the half-angle each cut makes with the outside edge.

FORMULA

$$\text{saw setting} = 180^\circ \div n$$

$$\text{corner angle } C = 180^\circ - 360^\circ \div n$$

n = number of sides. Check: $90^\circ - C \div 2$ gives the same saw setting.

MITER SETTINGS FOR POLYGON FRAMES				
Shape	Sides n	Corner angle	Cut to edge	Saw setting
Triangle	3	60°	30°	60°
Square	4	90°	45°	45°
Pentagon	5	108°	54°	36°
Hexagon	6	120°	60°	30°
Heptagon	7	128.57°	64.29°	25.71°
Octagon	8	135°	67.5°	22.5°
Nonagon	9	140°	70°	20°
Decagon	10	144°	72°	18°
Hendecagon	11	147.27°	73.64°	16.36°
Dodecagon	12	150°	75°	15°

WORKED EXAMPLE

An octagonal mirror frame.

1. $n = 8$
2. Saw setting: $180^\circ \div 8 = 22.5^\circ$ (most miter saws have a click-stop, called a **detent**, right at 22.5°).
3. Corner angle: $180^\circ - 360^\circ \div 8 = 135^\circ$. Each cut makes 67.5° with the edge.

Cut 16 ends (both ends of 8 pieces) at 22.5°, flipping the board between cuts so the long points land on the same edge.

COMMON MISTAKE

Triangle frames need a 60° setting, and many miter saws stop at 45°–50°. Check your saw’s maximum before you buy wood. Beyond it, mark the line with a protractor and cut it with a handsaw, or use a jig that holds the board at an angle to the fence.

2.3 Corners that aren’t 90°

Real walls are rarely perfect. A bay window may have 135° corners, a hexagonal room 120°, and an ordinary “square” corner can easily be 91° or 92°. The half-angle rule still works: measure the real corner, halve it, convert to a saw setting.

Measure on the side where the boards will sit. For baseboard in a room, that is the room side of the walls. A sliding bevel or a digital angle finder does this well (see 6.2).

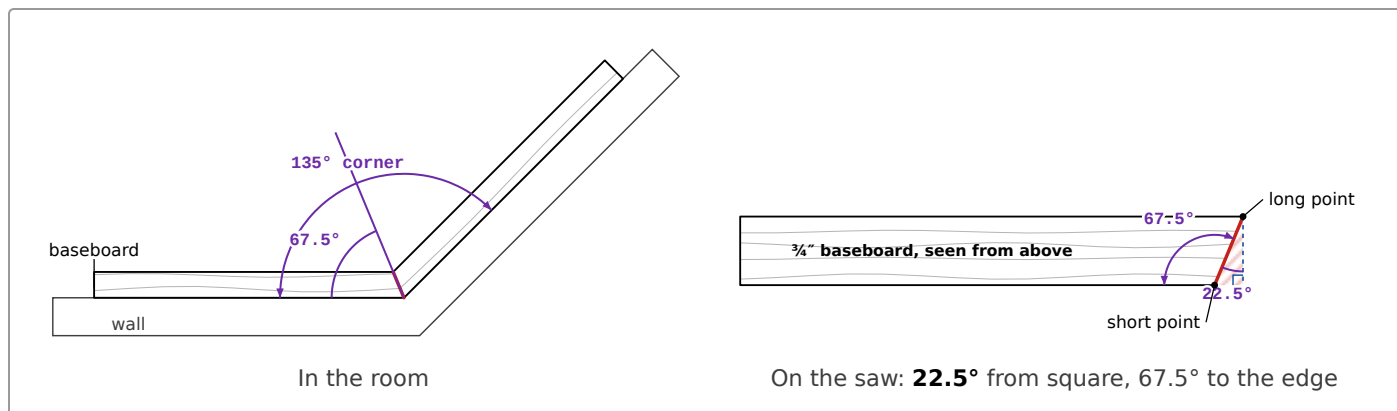


FIG 2.3 Baseboard meeting at a 135° bay-window corner, top view. The joint lies on the bisector (dashed purple). Right: one piece as it sits on the saw, standing against the fence.

FORMULA

$$\text{saw setting} = 90^\circ - (\text{measured corner} \div 2)$$

WORKED EXAMPLE

1. Bay corner measures 135°: $90^\circ - 135^\circ \div 2 = 90^\circ - 67.5^\circ = 22.5^\circ$
2. Hexagonal room corner 120°: $90^\circ - 60^\circ = 30^\circ$
3. A “square” corner that measures 92°: $90^\circ - 46^\circ = 44^\circ$

The out-of-square wall needs 44°, not 45°. Cutting at 45° would leave a gap you can see.

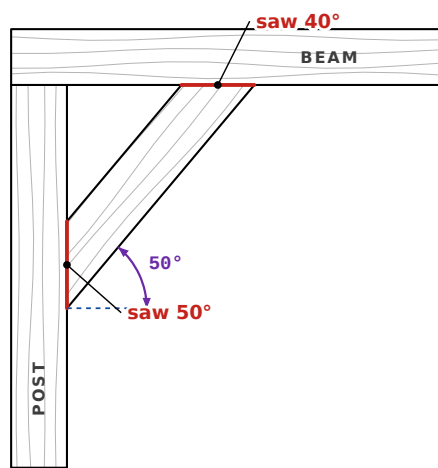
COMMON MISTAKE

Assuming a room corner is 90° because it looks square. A 2° error splits into 1° per piece, which opens about $\frac{1}{16}$ " on a $3\frac{1}{2}$ " board. Measure every corner; it takes seconds.

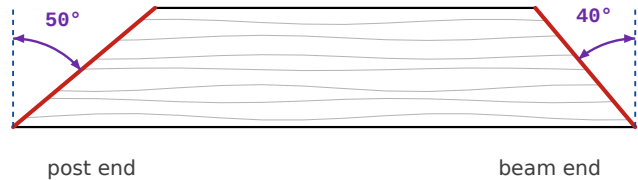
2.4 Braces, T-joints and boards landing on a slope

When a board ends against the *face* of another board (a T-joint, or a brace landing on a post), only the arriving board gets cut. Its end must lie flat against the surface it meets, so the cut line is parallel to that surface. There is nothing to share, so that board takes the *whole* angle, not half.

A diagonal **brace** between a post and a beam is the classic case. It has two ends, one against each member, and its two saw settings always add up to 90° when the post and beam are square to each other.



Installed



On the saw

FIG 2.4 A knee brace at 50° from horizontal. Left: installed. Right: the brace alone, laid flat as it would be on the saw, with each end's square line and saw setting.

FORMULA

end against horizontal surface: saw = $90^\circ - a$

end against vertical surface: saw = a

a = the brace's angle above horizontal. Both ends together: $(90^\circ - a) + a = 90^\circ$.

WORKED EXAMPLE

A brace that rises at 50° from horizontal.

1. Top end, flat against the underside of the beam: $90^\circ - 50^\circ = 40^\circ$
2. Bottom end, flat against the side of the post: 50°
3. Check: $40^\circ + 50^\circ = 90^\circ$ ✓

A 45° brace is the easy special case: 45° at both ends.

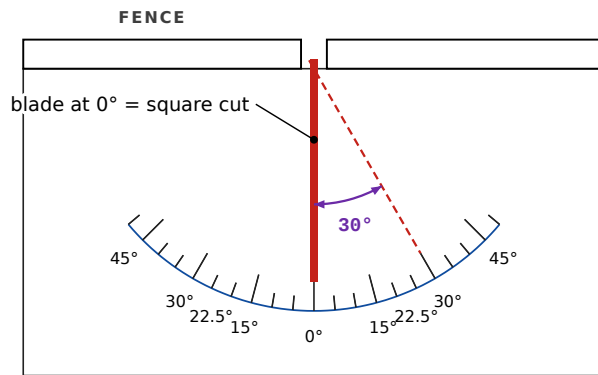
COMMON MISTAKE

Cutting both brace ends at the same setting because they “look alike.” Unless the brace is exactly 45°, one end needs a and the other $90^\circ - a$. Hold the brace in place and mark both ends before cutting, or label them “post” and “beam.”

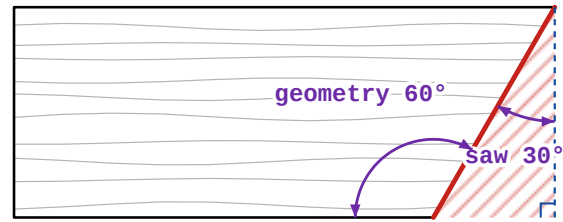
2.5 The gotcha: saws don't label angles like geometry does

In geometry, a square cut makes a 90° angle with the board's edge. A miter saw calls that same cut **0°**, because its scale counts how far the blade has swung *away from square*. Bevel scales work the same way: 0° is a vertical blade.

So there are two ways to describe any cut: the **geometric angle** (between the cut line and the edge of the board) and the **saw setting** (between the cut line and a square line). They always add up to 90°.



The saw scale: 0° is square



$$30^\circ + 60^\circ = 90^\circ$$

The same cut, two names

FIG 2.5 Left: a miter saw table seen from above. The blade at 0° is square to the fence; the ghost blade is swung to 30°. Right: the same 30° cut drawn on the board. Measured from the edge it is 60°.

FORMULA

saw setting = $90^\circ - \text{geometric angle}$

geometric angle = $90^\circ - \text{saw setting}$

CONVERTING BETWEEN THE TWO SCALES

Saw setting	Geometric angle (to edge)	Where you see it
0°	90°	square crosscut
15°	75°	shallow miters
22.5°	67.5°	octagon
30°	60°	hexagon
45°	45°	square frame
60°	30°	triangle

WORKED EXAMPLE

A plan says “cut the end at 70° to the edge.”

$$1. 90^\circ - 70^\circ = 20^\circ$$

Set the miter saw to 20°. Setting it to 70° is impossible on most saws, which is often the hint that you have the wrong scale.

COMMON MISTAKE

Reading an angle off a protractor (geometry) and dialing that number straight into the saw. Before setting the saw, ask: “Is 0 a square cut here?” If it is, you are on the saw scale.

WHAT AN ANGLED CUT DOES TO LENGTH

Long-to-short offset · inside vs outside length · bevel lengths · offset tables · no-calculator shortcuts

3.1 The offset between long and short point

Look at the end of a mitered board from above. The cut line, the square line, and the edge of the board form a right triangle (a triangle with one 90° corner). The board's width is one side. The distance you slide along the edge from the short point to the long point is the other side. Call it the **offset**.

The steeper the saw setting, the longer that slide. The **tangent** (tan) of an angle is simply the ratio “side across from the angle ÷ side next to it,” so it converts width into offset. Any phone calculator has a tan button; make sure it is in degrees (DEG), not radians.

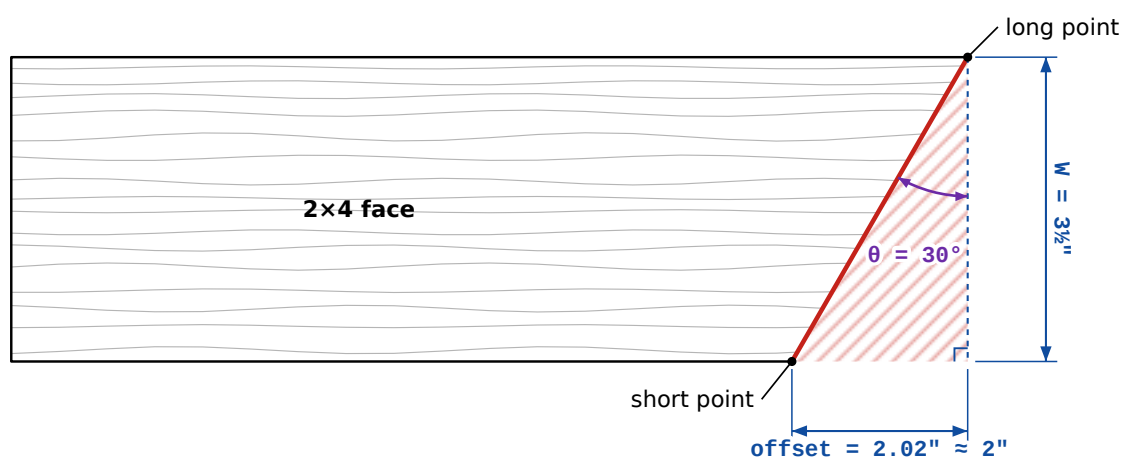


FIG 3.1 The end of a 2×4 face cut at a 30° saw setting. The hatched triangle is the waste; its two short sides are the board width W and the offset.

FORMULA

$$\text{offset} = W \times \tan(\theta)$$

W = width of the surface being cut across (3½" on a 2×4 face). θ = saw setting (measured from square).

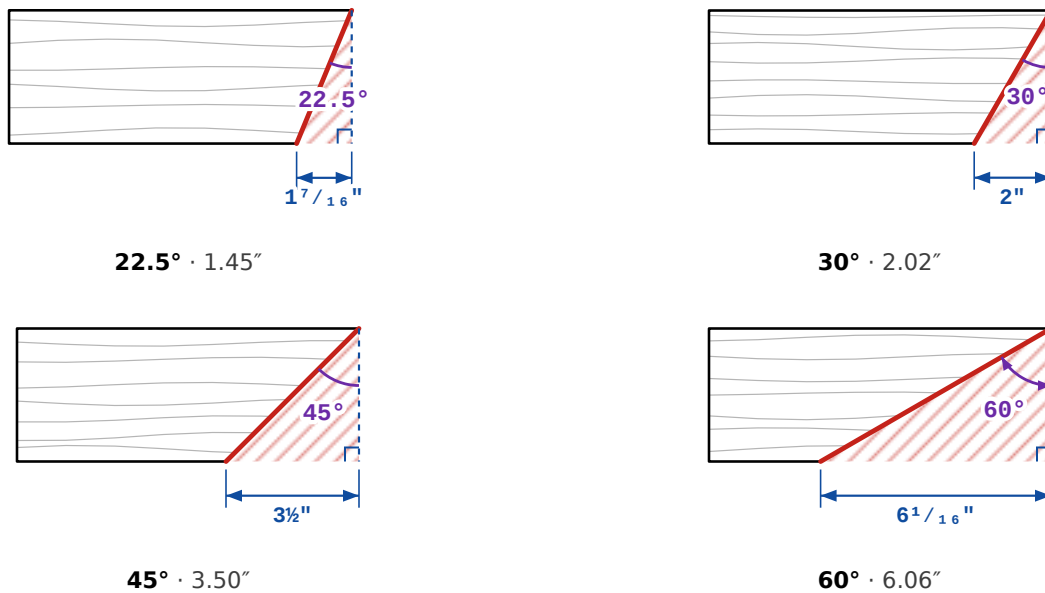


FIG 3.2 The four common angles on a 3½" 2×4 face, all drawn at the same scale.

WORKED EXAMPLE

2×4 face, W = 3.5":

1. 22.5°: $3.5 \times 0.4142 = 1.45" \approx 1\frac{7}{16}"$
2. 30°: $3.5 \times 0.5774 = 2.02" \approx 2"$
3. 45°: $3.5 \times 1.0000 = 3.50" = 3\frac{1}{2}"$
4. 60°: $3.5 \times 1.7321 = 6.06" \approx 6\frac{1}{16}"$

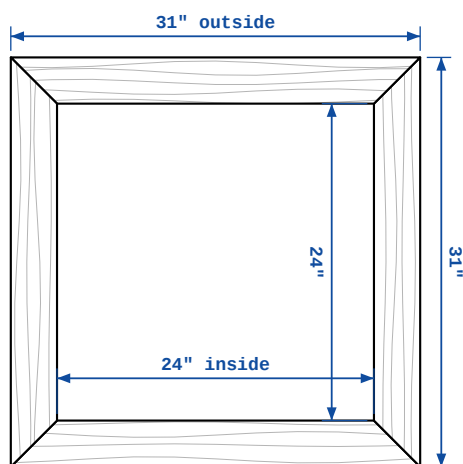
Going from 45° to 60° almost doubles the offset. Offsets grow quickly at steep angles.

COMMON MISTAKE

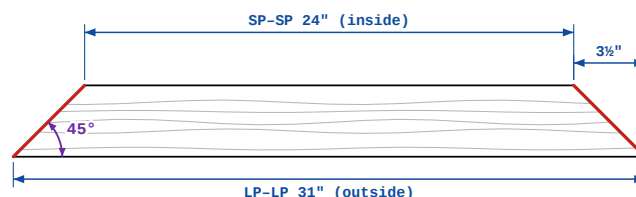
Using the geometric angle (60° to the edge) where the formula wants the saw setting (30° from square). That swaps $\tan 30^\circ$ for $\tan 60^\circ$ and triples the offset. If your answer is bigger than the board is wide at anything under 45°, you've used the wrong angle.

3.2 Inside length vs outside length

A mitered frame piece has two lengths: the **outside length** (long point to long point) and the **inside length** (short point to short point). The inside length is what you see around the opening, like the picture or the mirror. Each mitered end adds one offset, so a piece mitered at both ends is two offsets longer on the outside.



Assembled frame



One piece: $24 + 3\frac{1}{2} + 3\frac{1}{2} = 31$

FIG 3.3 A square frame with a 24" opening, made from 2×4s laid flat. Left: assembled. Right: one piece pulled out, with both lengths dimensioned.

FORMULA

$$\text{outside (LP-LP)} = \text{inside (SP-SP)} + 2 \times W \times \tan(\theta)$$

$$\text{inside} = \text{outside} - 2 \times W \times \tan(\theta)$$

If the two ends are cut at different angles θ_1 and θ_2 (long points on the same edge), use $w \times \tan \theta_1 + W \times \tan \theta_2$ instead of $2 \times W \times \tan \theta$.

WORKED EXAMPLE

"I need a square frame whose opening is 24" × 24", from 2×4s laid flat."

1. Square frame → saw at 45°. Offset: $3.5 \times \tan 45^\circ = 3.5"$
2. Long point length: $24 + 2 \times 3.5 = 31"$
3. The frame's outside size is also 31" × 31".

Cut four pieces at 31" LP-LP. Second example, a hexagon with 12" inside sides: offset $3.5 \times \tan 30^\circ = 2.02"$, so each piece is $12 + 4.04 = 16.04" \approx 16\frac{1}{16}"$ LP-LP.

COMMON MISTAKE

Cutting pieces to the opening size (24") as if the corners were square. The frame then comes out 7" too small inside. Decide first whether your number is an inside or an outside dimension, and write LP or SP next to it.

3.3 The same idea for a bevel

A bevel slopes through the thickness instead of across the face. The triangle is the same, it just stands on its edge. So the formula is identical; only W changes: use the thickness the blade passes through. For a 2×4 lying flat and beveled, that is $1\frac{1}{2}"$.

Typical use: a four-sided box or planter frame where the corners are beveled so no end grain shows on the outside.

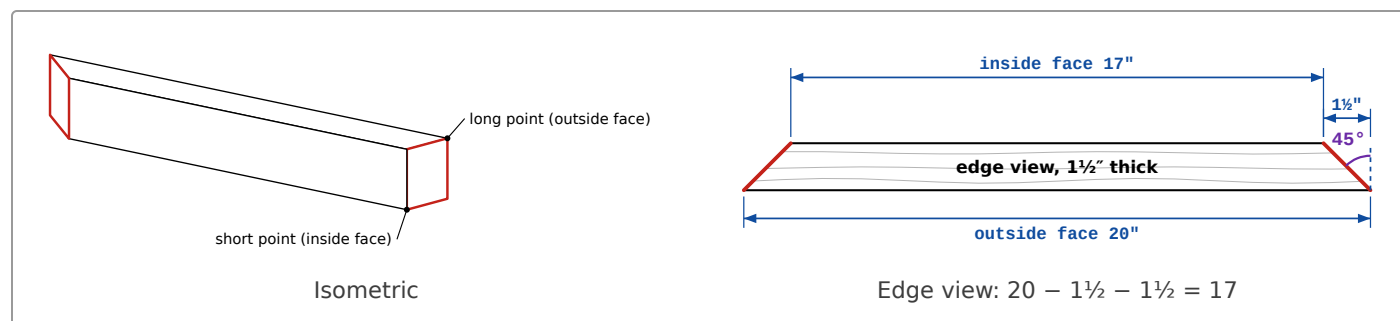


FIG 3.4 One side of a 2×4 box frame, beveled 45° at both ends through the 1½" thickness. Left: isometric. Right: edge view with both lengths.

FORMULA

$$\text{offset} = T \times \tan(\text{bevel})$$

$$\text{outer face} = \text{inner face} + 2 \times T \times \tan(\text{bevel})$$

T = thickness being cut through (1½" for a 2×4, ¾" for a 1×4).

WORKED EXAMPLE

A square 2×4 frame (boards standing on edge) that must be 20" across on the outside, corners beveled at 45°.

1. Offset: $1.5 \times \tan 45^\circ = 1.5"$
2. Outer face (long point to long point): 20"
3. Inner face: $20 - 2 \times 1.5 = 17"$

An octagonal column wrap at 22.5° bevel on the same 2×4: offset $1.5 \times 0.4142 = 0.62" \approx \frac{5}{8}"$ per end.

COMMON MISTAKE

Using the face width (3½") for a bevel. The blade passes through 1½" of wood when beveling a flat 2×4, so the offset is much smaller. Look at what the blade actually travels through.

3.4 Offset table for common lumber

Nominal sizes (what the store calls it) and actual sizes (what you measure) differ. The table uses actual sizes: 2×4 = 1½" × 3½", 2×6 = 1½" × 5½", 1×4 = ¾" × 3½". Face columns are for miters across the face; thickness columns are for bevels.

OFFSET FROM SHORT POINT TO LONG POINT (DECIMAL INCHES, THEN NEAREST $\frac{1}{16}$ ")						
Saw θ	$\tan \theta$	2×4 face 3½"	2×4 thick 1½"	2×6 face 5½"	1×4 face 3½"	1×4 thick ¾"
5°	0.0875	0.31 $\frac{5}{16}$ "	0.13 $\frac{1}{8}$ "	0.48 $\frac{1}{2}$ "	0.31 $\frac{5}{16}$ "	0.07 $\frac{1}{16}$ "
10°	0.1763	0.62 $\frac{5}{8}$ "	0.26 $\frac{1}{4}$ "	0.97 1"	0.62 $\frac{5}{8}$ "	0.13 $\frac{1}{8}$ "
15°	0.2679	0.94 $\frac{15}{16}$ "	0.40 $\frac{3}{8}$ "	1.47 $1\frac{1}{2}$ "	0.94 $\frac{15}{16}$ "	0.20 $\frac{3}{16}$ "
22.5°	0.4142	1.45 $1\frac{7}{16}$ "	0.62 $\frac{5}{8}$ "	2.28 $2\frac{1}{4}$ "	1.45 $1\frac{7}{16}$ "	0.31 $\frac{5}{16}$ "
30°	0.5774	2.02 2"	0.87 $\frac{7}{8}$ "	3.18 $3\frac{1}{16}$ "	2.02 2"	0.43 $\frac{7}{16}$ "
36°	0.7265	2.54 $2\frac{9}{16}$ "	1.09 $1\frac{1}{16}$ "	4.00 4"	2.54 $2\frac{9}{16}$ "	0.54 $\frac{9}{16}$ "
45°	1.0000	3.50 $3\frac{1}{2}$ "	1.50 $1\frac{1}{2}$ "	5.50 $5\frac{1}{2}$ "	3.50 $3\frac{1}{2}$ "	0.75 $\frac{3}{4}$ "
60°	1.7321	6.06 $6\frac{1}{16}$ "	2.60 $2\frac{5}{8}$ "	9.53 $9\frac{1}{2}$ "	6.06 $6\frac{1}{16}$ "	1.30 $1\frac{5}{16}$ "

COMMON MISTAKE

Using nominal sizes (4", 6") in the formula. A 45° miter on a "2×6" moves the long point 5½", not 6".

3.5 Estimating without a calculator

A handful of ratios cover most jobs. Memorise 45° and the "one-sixteenth per degree" rule and you can sanity-check almost any cut list in your head.

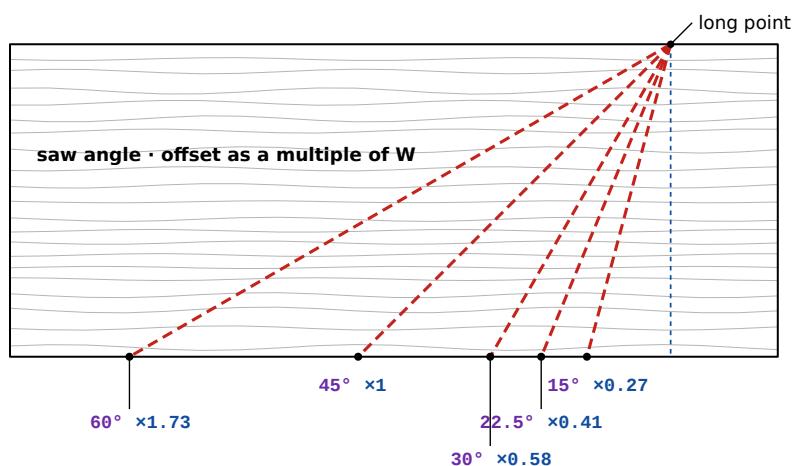


FIG 3.5 Every ray starts at the long point. Where it hits the far edge gives the offset as a multiple of the board's width W.

RULES OF THUMB

- **45°:** offset = width. A 3½" board moves 3½".
- **30°:** a bit over half the width ($\times 0.58$). 2×4: about 2".
- **22.5°:** a bit over two-fifths ($\times 0.41$). 2×4: about $1\frac{7}{16}$ ".
- **15°:** about a quarter ($\times 0.27$). 2×4: about $\frac{15}{16}$ ".
- **60°:** about $1\frac{3}{4}$ widths ($\times 1.73$). 2×4: about 6".
- **Small angles:** each degree adds about $W \div 57$. On a 3½" face that is almost exactly $\frac{1}{16}$ " per degree, and it stays within $\frac{1}{32}$ " up to about 20°.

WORKED EXAMPLE

A 10° miter on a 2×4 face, no calculator.

$$1. 10^\circ \times \frac{1}{16}'' = \frac{5}{8}'' \text{ (exact answer } 0.617'')$$

Close enough to mark. For anything cut on a saw, you don't need the offset at all: set the angle and measure the long-point length.

COMMON MISTAKE

Stretching the per-degree rule to steep angles. At 45° it predicts 2.8" instead of 3.5". Past about 20°, switch to the named ratios above.

SLOPES, RISE AND RUN

Pitch (X-in-12) ↔ degrees · plumb cuts · level cuts · rafter tails and brace ends

4.1 Pitch: slope as “inches per foot”

Builders often describe a slope by how much it climbs over a horizontal foot instead of in degrees. The climb is the **rise**, the horizontal distance is the **run**, and “6-in-12” **pitch** means 6” of rise for every 12” of run. On roof drawings it is shown as a small triangle with 12 on the flat side.

Pitch is easy to measure with a level and a tape, and it scales: a 6-in-12 roof rises 3” over 6” and 5 feet over 10 feet.

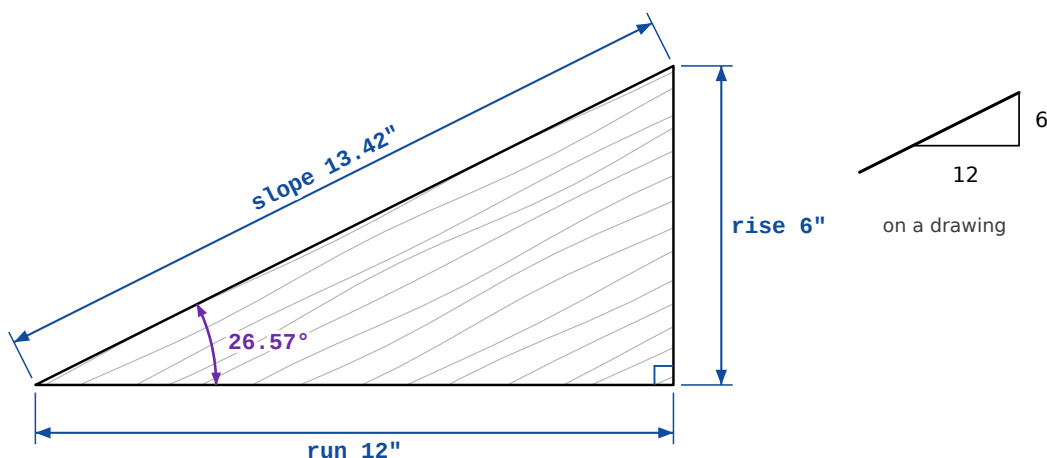


FIG 4.1 A 6-in-12 slope. The sloped side is the length of rafter needed per 12” of run.

FORMULA

$$\text{angle} = \arctan(\text{rise} \div 12)$$

$$\text{rise per 12} = 12 \times \tan(\text{angle})$$

$$\text{sloped length} = \text{run} \times \sqrt{1 + (\text{rise} \div 12)^2}$$

arctan (also written $\tan^{-1}$ or atan) is the calculator button that turns a ratio back into an angle. The square-root factor is the “slope factor.”

PITCH TO DEGREES				
Pitch	Degrees	Plumb cut (saw)	Level cut (saw)	Slope factor
1-in-12	4.76°	4.8°	85.2°	1.003
2-in-12	9.46°	9.5°	80.5°	1.014
3-in-12	14.04°	14°	76°	1.031
4-in-12	18.43°	18.4°	71.6°	1.054
5-in-12	22.62°	22.6°	67.4°	1.083
6-in-12	26.57°	26.6°	63.4°	1.118
7-in-12	30.26°	30.3°	59.7°	1.158
8-in-12	33.69°	33.7°	56.3°	1.202
9-in-12	36.87°	36.9°	53.1°	1.250
10-in-12	39.81°	39.8°	50.2°	1.302
11-in-12	42.51°	42.5°	47.5°	1.357
12-in-12	45°	45°	45°	1.414

WORKED EXAMPLE

1. 6-in-12 to degrees: $\arctan(6 \div 12) = \arctan(0.5) = 26.57^\circ$
2. 30° to pitch: $12 \times \tan 30^\circ = 6.93 \rightarrow$ about 7-in-12
3. Rafter over a 10 ft run at 6-in-12: $10 \times \sqrt{1.25} = 11.18 \text{ ft} = 11' 2\frac{3}{16}"$

COMMON MISTAKE

Treating pitch as a percentage or as degrees. 6-in-12 is 50% grade and 26.57° , not 6° and not 60%. Always keep the “in-12”.

4.2 Cutting sloped pieces: plumb and level cuts

A board lying on a slope usually needs its end cut **plumb** (truly vertical, so it looks right to the eye or sits flat against a vertical fascia board) or **level** (truly horizontal, so it sits flat on something or takes a soffit board). The end of a rafter that sticks out past the wall is the **rafter tail**.

This is the brace rule from 2.4 again: a plumb cut lands on something vertical, a level cut on something horizontal.

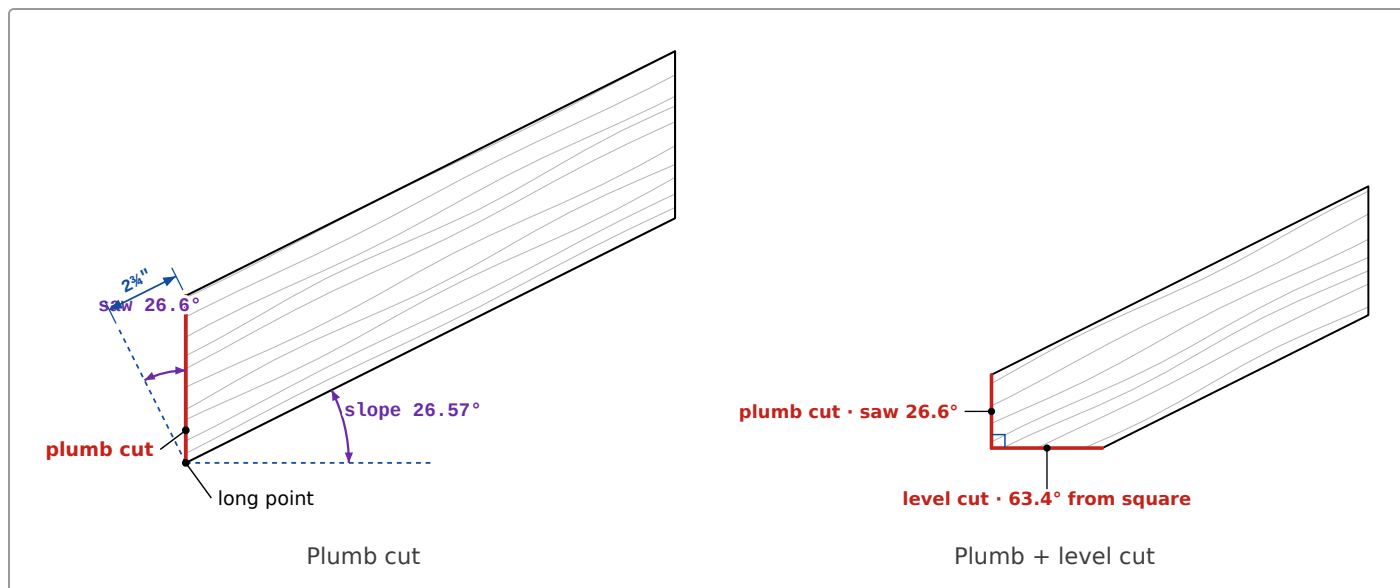


FIG 4.2 A 2×6 rafter tail at 6-in-12. Left: plumb-cut tail. Right: plumb cut plus a level cut underneath.

FORMULA

plumb cut: saw setting = slope angle

level cut: saw setting = $90^\circ - \text{slope angle}$

plumb-cut offset = $W \times \text{rise} \div 12$

The last line is $W \times \tan(\text{slope})$ written in pitch terms, because $\tan(\text{slope})$ is just $\text{rise} \div 12$.

WORKED EXAMPLE

2×6 rafter ($W = 5\frac{1}{2}"$) at 6-in-12:

1. Slope angle: 26.57° , so the plumb cut is 26.6° on the saw (a hair over the 26.5° many people use).
2. Plumb-cut offset: $5.5 \times 6 \div 12 = 2.75" = 2\frac{3}{4}"$
3. Level cut: $90^\circ - 26.57^\circ = 63.4^\circ$ from square. That is past most saws, so mark it with a square and cut it with a circular saw or handsaw.

COMMON MISTAKE

Marking the plumb line square to the rafter instead of truly vertical. Lay a speed square or level against the rafter *in position*, or use the pitch scale on a speed square, so the line is plumb in the finished building.

COMPOUND CUTS (INTRO)

Splayed boxes · crown molding · tapered planters · miter and bevel settings from the tilt

5.1 What they are and when you need them

You need a compound cut when the boards meeting at a corner are *tilted*, not standing straight up or lying flat. Examples: a planter whose sides lean outward, a tray with sloped sides, a hopper, and crown molding, which sits tilted between wall and ceiling.

The surprise is that neither saw setting equals the angle you see. A four-sided planter still turns 90° at each corner, but because the sides lean, the joint line is no longer a simple 45° across the face.

The miter and bevel have to share the job, and how they share it depends on the tilt.

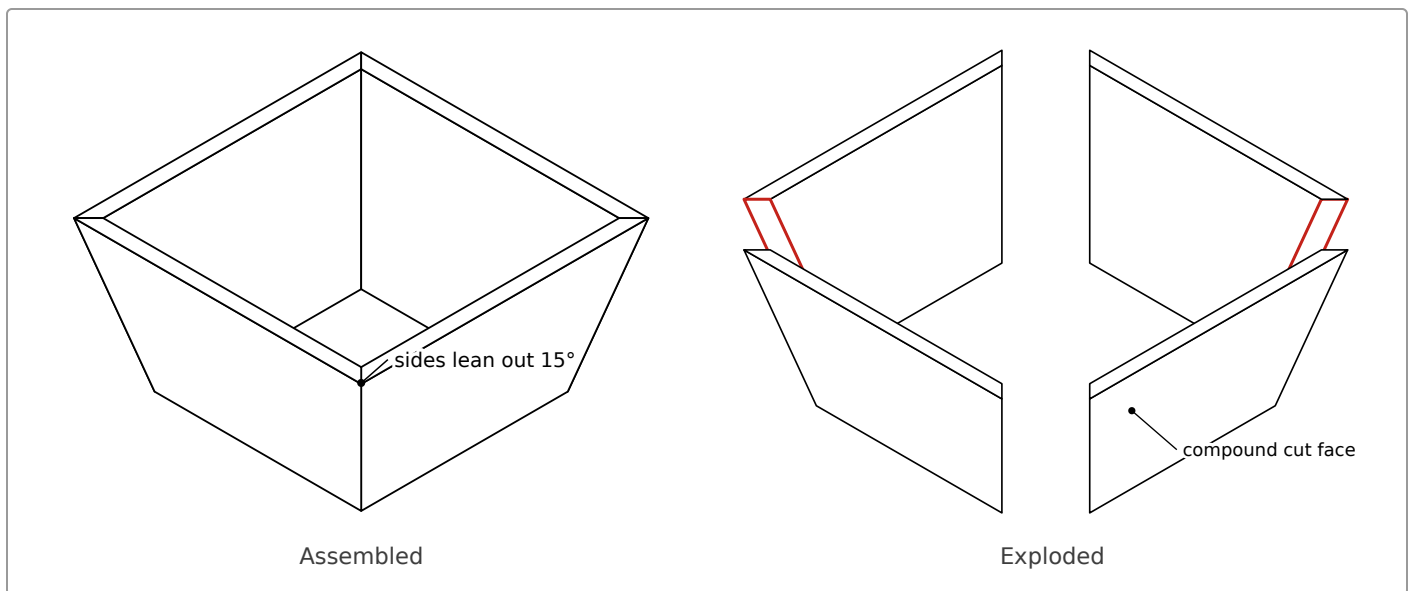


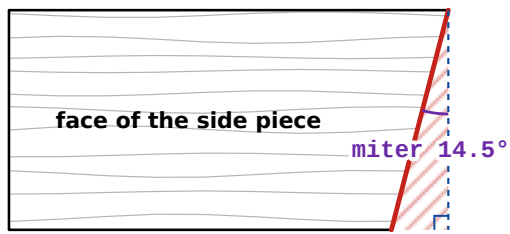
FIG 5.1 A four-sided planter with sides leaning out 15° from vertical. Left: assembled. Right: exploded so you can see the compound cut faces (red outlines).

COMMON MISTAKE

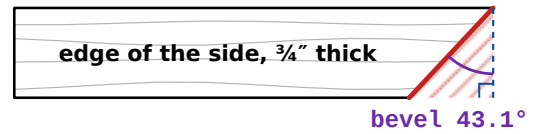
Cutting a splayed box at 45° miter and 15° bevel because “it’s square with a 15° lean.” The corners will not close. Use the formulas below; the settings are 14.5° and 43.1° here.

5.2 Working out the miter and bevel settings

Two limits give the intuition. If the sides stand straight up (tilt 0°), a square box is cut with a pure 45° bevel. If the sides lie flat (tilt 90°), it is a picture frame with a pure 45° miter. For tilts in between, the miter grows from 0° toward 45° and the bevel shrinks from 45° toward 0° .



From above: **14.5°** miter



From the front: **43.1°** bevel

FIG 5.2 The 15° planter side as the saw sees it, lying face-down on the table. Left: from above, the miter setting. Right: from the front, the bevel (blade tilt).

FORMULA

$$A = 180^\circ \div n \text{ (or } 90^\circ - C \div 2 \text{ for any corner } C)$$

$$\text{miter} = \arctan(\sin(S) \times \tan(A))$$

$$\text{bevel} = \arcsin(\cos(S) \times \sin(A))$$

S = tilt of the sides from vertical. A = the flat-frame saw setting for that corner. Board lies face-down on the saw table. For crown molding, S is the angle between the molding's back and the wall: 38° for common "52/38" crown, 45° for "45/45" crown.

WORKED EXAMPLE

Four-sided planter, sides tilted 15° from vertical.

$$1. A = 180^\circ \div 4 = 45^\circ$$

$$2. \text{Miter: } \arctan(\sin 15^\circ \times \tan 45^\circ) = \arctan(0.2588 \times 1) = 14.51^\circ$$

$$3. \text{Bevel: } \arcsin(\cos 15^\circ \times \sin 45^\circ) = \arcsin(0.9659 \times 0.7071) = \arcsin(0.6830) = 43.08^\circ$$

Set the saw to about 14.5° miter and 43° bevel. A 45° tilt gives the well-known pair 35.26° and 30°.

COMPOUND SETTINGS (BOARD FACE-DOWN ON THE SAW)				
Project	Sides	Tilt from vertical	Miter	Bevel
Box / planter	4	5°	4.98°	44.78°
Box / planter	4	10°	9.85°	44.14°
Box / planter	4	15°	14.51°	43.08°
Box / planter	4	20°	18.88°	41.64°
Box / planter	4	25°	22.91°	39.86°
Box / planter	4	30°	26.57°	37.76°
Box / planter	4	45°	35.26°	30°
Hexagon planter	6	15°	8.5°	28.88°
Hexagon planter	6	30°	16.1°	25.66°
Octagon planter	8	15°	6.12°	21.69°
Octagon planter	8	30°	11.7°	19.35°
Crown 52/38, 90° corner	4	38° spring	31.62°	33.86°
Crown 45/45, 90° corner	4	45° spring	35.26°	30°

COMMON MISTAKE

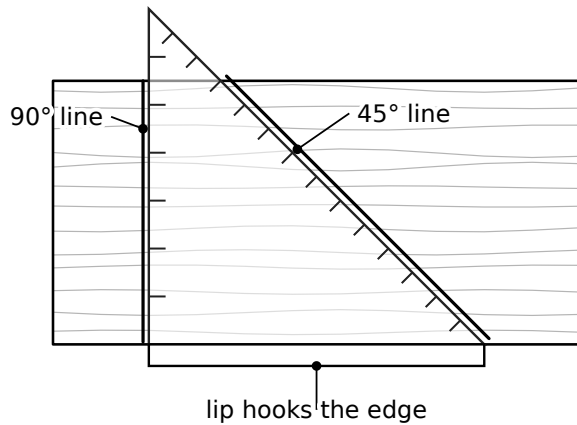
Mixing up which way the sides are measured. This table uses tilt *from vertical*. Some charts use the slope from horizontal (90° minus our number). If a chart gives you a bevel above 45° for a square box, it is using the other convention.

MEASURING, MARKING AND CHECKING

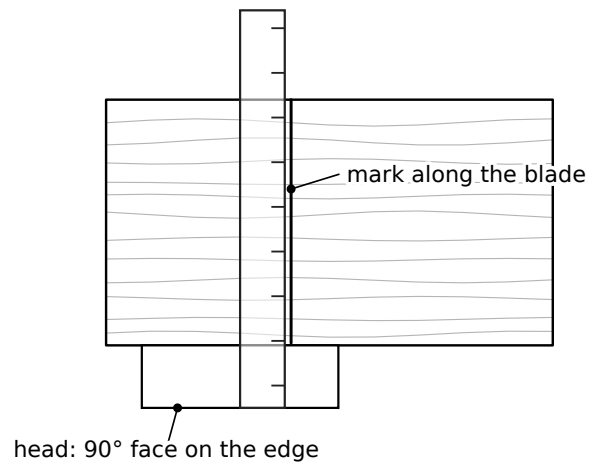
Speed square · combination square · sliding bevel · protractor · transferring angles · kerf and waste side · test cuts and error build-up

6.1 The four layout tools

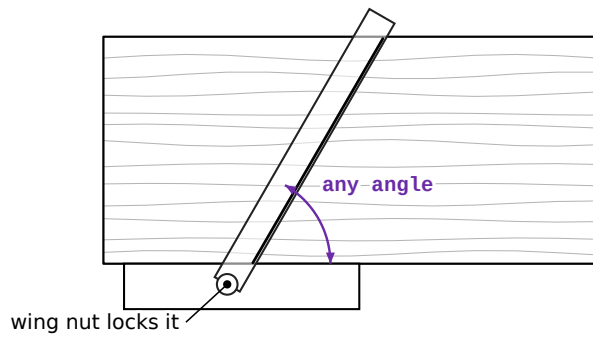
- **Speed square** (rafter square): a thick triangle with a lip. Hook the lip on the edge; the leg draws a 90° line and the long side draws a 45° line. For other angles, hold the corner marked *PIVOT* on the edge and swing the square until the edge of the board lines up with the number you want on the degree scale. Most speed-square degree scales read like the saw (0 = square); check yours by drawing the 0 line and comparing it with the 90° leg.
- **Combination square**: a steel ruler that slides through a cast head. The head has a 90° face and a 45° face. It only does those two angles, but very accurately, and it doubles as a depth gauge and marking gauge.
- **Sliding bevel** (bevel gauge): a handle and a blade locked by a wing nut. It has no scale; it *copies* an angle from a corner, a drawing or a protractor and carries it to the wood or the saw.
- **Protractor**: a half-circle marked 0° to 180°. Put the center mark on your point with the base along the edge and read where your line crosses the scale. It reads geometry (90° = square), so convert before setting the saw.



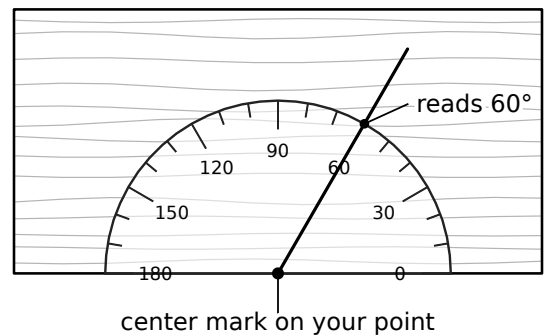
Speed square



Combination square



Sliding bevel



Protractor

FIG 6.1 Each tool sitting on a board (top view). Pencil lines are dark; each tool's reference face presses against the board's edge.

COMMON MISTAKE

Letting the tool's fence lift off the edge while you mark. Press the lip or head firmly against the edge with one hand and draw with the other, pulling the pencil toward you with its point tight against the blade.

6.2 Copying an angle from a real corner to the saw

1. Set a sliding bevel (or digital angle finder) into the corner, handle against one wall and blade against the other. Lock it.
2. Read the angle by laying the bevel on a protractor, or skip numbers and halve it on paper (below).
3. Saw setting = $90^\circ - \text{corner} \div 2$.
4. Cut two scrap pieces and hold them in the corner before cutting good wood.

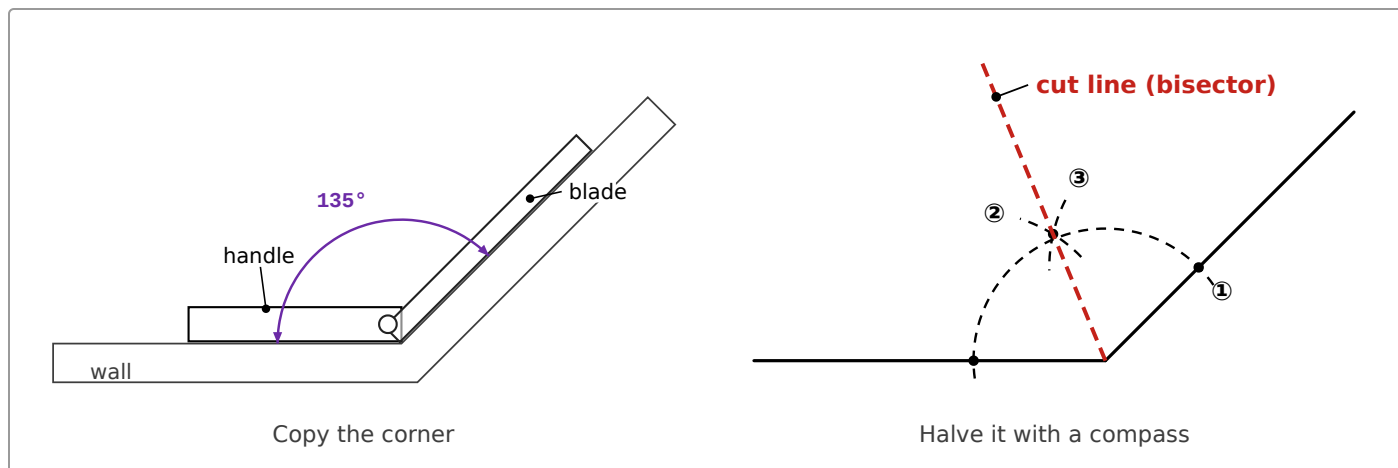


FIG 6.2 Left: a sliding bevel copying a 135° corner. Right: halving it without numbers. Trace the angle, swing arc ① from the corner, then equal arcs ② and ③ from where ① crosses the lines. The line through their crossing is the cut line.

WORKED EXAMPLE

The bevel, laid on a protractor, reads 131° .

$$1. 90^\circ - 131^\circ \div 2 = 90^\circ - 65.5^\circ = 24.5^\circ$$

Set the saw to 24.5° . Many saws let you override the 22.5° click-stop and lock anywhere in between.

COMMON MISTAKE

Measuring the corner at the floor, where drywall mud and tape build up. Measure at the height the trim will actually sit, and check a second spot along the corner.

6.3 Measuring to the long point and cutting on the waste side

Hook the tape on the long point, mark the length on the same edge with a small “V” (a **crow’s foot**) whose tip is exactly on the number, and put an X on the waste side. Then line the blade up so its teeth just touch the tip of the V from the waste side. The kerf then comes out of the waste, and the keeper is exactly the length you marked.

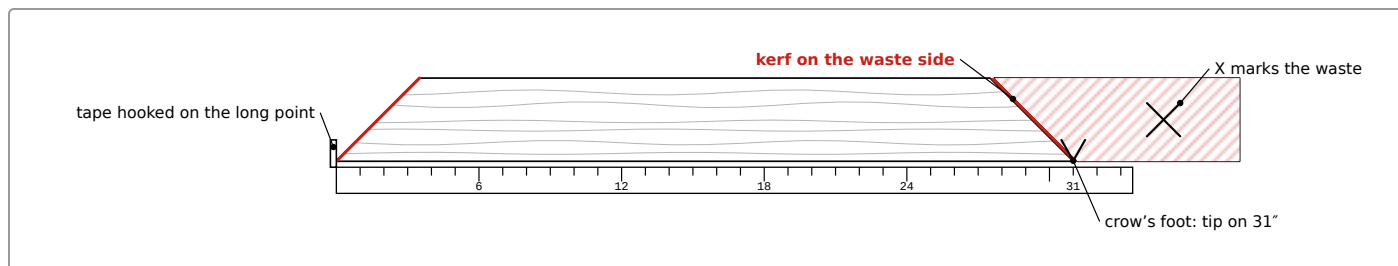


FIG 6.3 Marking a 31" LP-LP frame piece. The tape hooks the left long point; the kerf (red, $\frac{1}{8}$ ") sits just to the right of the mark.

WORKED EXAMPLE

Four 31" frame pieces cut with the blade centered on the line ($\frac{1}{8}$ " kerf):

1. Each second cut shortens its piece by half a kerf: $31" - \frac{1}{16}" = 30\frac{15}{16}"$

On a square frame that is only $\frac{1}{16}$ ", but if two opposite sides come out different lengths, the frame won't close square.

COMMON MISTAKE

Cutting pairs to length separately. Clamp opposite sides together (or use a **stop block**, a block clamped to the saw fence at the right distance) so pairs come out identical, even if they are a hair off the plan.

6.4 Test cuts and how small errors add up

A saw that is half a degree off looks perfect on a single cut. The trouble is that a frame adds up every cut. An octagon has 16 cut ends, and all their errors pile into the last joint you close.

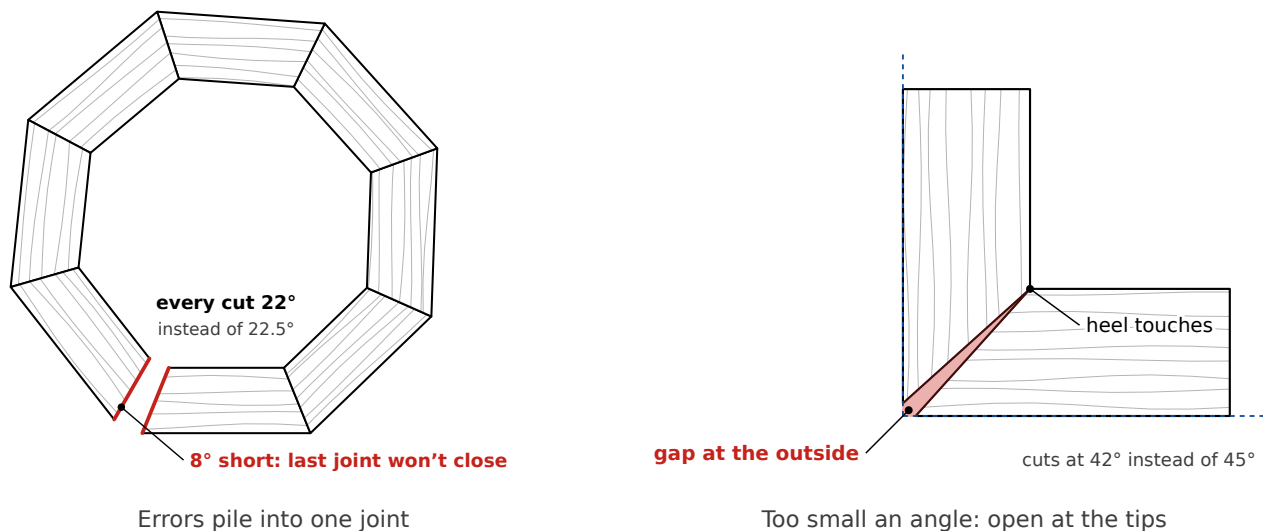


FIG 6.4 Left: an octagon dry-fitted with every cut 0.5° shy of 22.5° . Joints 1–7 close tight, and the missing 8° all shows up at the last joint. Right: a single corner cut 3° shy (exaggerated) opens at the outside tips.

FORMULA

total error = (number of cut ends) $\times$ (error per cut)

gap at the joint $\approx W \times \tan(\text{total error})$

WORKED EXAMPLE

1. Octagon, 16 cut ends, each 0.5° off: $16 \times 0.5^\circ = 8^\circ$
2. Gap across a $3\frac{1}{2}"$ joint: $3.5 \times \tan 8^\circ = 0.49"$, about $\frac{1}{2}"$
3. Square frame, 8 cut ends, each 0.5° off: $4^\circ \rightarrow 3.5 \times \tan 4^\circ = 0.24"$, about $\frac{1}{4}"$

To correct from a dry fit, divide the gap angle by the number of cut ends: an 8° gap on an octagon means each cut is $8^\circ \div 16 = 0.5^\circ$ off.

TEST-CUT ROUTINE

- Cut two scrap pieces, put them together, and check the corner with a square. Any error you see is *double* the per-cut error.
- Gap at the outside tips: the saw angle is too *small*. Gap at the inside (heel): too *large*.
- For polygons, cut a full set of short test pieces and dry-fit the whole ring before cutting good wood.

COMMON MISTAKE

Trusting the saw's detents. Click-stops wear and can be off by half a degree from the factory. Check 90° and 45° with a combination square after any bump or blade change.

QUICK-REFERENCE CHEAT SHEET

Every formula and table from this guide on one sheet, plus a small calculator

KEY FORMULAS

SAW ↔ GEOMETRY

saw = $90^\circ - \text{geometric}$

CORNER (BOTH PIECES CUT)

saw = $90^\circ - C \div 2$

POLYGON, N SIDES

saw = $180^\circ \div n$

BRACE ENDS

horizontal end $90^\circ - a \cdot \text{vertical end } a$

OFFSET (MITER)

$W \times \tan \theta$

OFFSET (BEVEL)

$T \times \tan \theta$

FRAME PIECE

$LP-LP = SP-SP + 2 W \tan \theta$

PITCH → DEGREES

$\arctan(\text{rise} \div 12)$

DEGREES → PITCH

$12 \times \tan(\text{angle})$

SLOPE LENGTH

$\text{run} \times \sqrt{(1 + (\text{rise} \div 12)^2)}$

PLUMB / LEVEL CUT

slope $\cdot 90^\circ - \text{slope}$

COMPOUND (TILT S)

miter = $\text{atan}(\sin S \cdot \tan A)$

bevel = $\text{asin}(\cos S \cdot \sin A)$

KERF

stock = $\Sigma \text{ lengths} + \text{cuts} \times \text{kerf}$

ERROR BUILD-UP

gap $\approx W \times \tan(\text{ends} \times \text{error})$

POLYGON FRAMES

n	Corner	Saw
3 • Triangle	60°	60°
4 • Square	90°	45°
5 • Pentagon	108°	36°
6 • Hexagon	120°	30°
7 • Heptagon	128.57°	25.71°
8 • Octagon	135°	22.5°
9 • Nonagon	140°	20°
10 • Decagon	144°	18°
11 • Hendecagon	147.27°	16.36°
12 • Dodecagon	150°	15°

2x4 OFFSETS

Saw θ	Face 3½"	Bevel 1½"
5°	0.31 $\frac{5}{16}$ "	0.13 $\frac{1}{8}$ "
10°	0.62 $\frac{5}{8}$ "	0.26 $\frac{1}{4}$ "
15°	0.94 $\frac{15}{16}$ "	0.40 $\frac{3}{8}$ "
22.5°	1.45 $1\frac{7}{16}$ "	0.62 $\frac{5}{8}$ "
30°	2.02 2"	0.87 $\frac{7}{8}$ "
36°	2.54 $2\frac{9}{16}$ "	1.09 $1\frac{1}{16}$ "
45°	3.50 $3\frac{1}{2}$ "	1.50 $1\frac{1}{2}$ "
60°	6.06 $6\frac{1}{16}$ "	2.60 $2\frac{5}{8}$ "

PITCH → DEGREES		
Pitch	Degrees	Factor
1/12	4.76°	1.003
2/12	9.46°	1.014
3/12	14.04°	1.031
4/12	18.43°	1.054
5/12	22.62°	1.083
6/12	26.57°	1.118
7/12	30.26°	1.158
8/12	33.69°	1.202
9/12	36.87°	1.250
10/12	39.81°	1.302
11/12	42.51°	1.357
12/12	45°	1.414

COMPOUND (SQUARE BOX)		
Tilt	Miter	Bevel
0°	0°	45°
5°	4.98°	44.78°
10°	9.85°	44.14°
15°	14.51°	43.08°
20°	18.88°	41.64°
25°	22.91°	39.86°
30°	26.57°	37.76°
35°	29.84°	35.4°
40°	32.73°	32.8°
45°	35.26°	30°
60°	40.89°	20.7°

The Angled-Cut Handbook · first edition. All diagrams are generated in the page from the same formulas the text uses, so they are to scale. Lumber sizes are North American dimensional lumber (actual size). Always check a new setup on scrap first.